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User feedback on potential obstacles and error conditions detected by autonomous mobile robots

Abstract

A mobile computing device includes a user input device and a controller. The user input device includes a display, and the controller is operably connected to the user input device and configured to execute instructions to perform operations. The operations include presenting on the display, information about one or more areas that were not cleaned by an autonomous cleaning robot during a first mission. The operations further include transmitting data corresponding to a user-selected subset of the one or more areas to cause the robot to clean the user-selected subset during a second mission.

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Background/Summary

TECHNICAL FIELD

(1) This specification relates to the acquisition and use of user feedback on obstacles detected by autonomous mobile robots and related systems and methods.

BACKGROUND

(2) Autonomous mobile robots include autonomous cleaning robots that autonomously perform cleaning tasks within an environment, e.g., a home. Many kinds of cleaning robots are autonomous to some degree and in different ways. A cleaning robot can include a controller configured to autonomously navigate the robot about an environment such that the robot can ingest debris as it moves. The cleaning robot can include a sensor for avoiding obstacles in the environment.

SUMMARY

(3) An autonomous mobile robot can detect potential obstacles and/or error conditions while operating in an environment (e.g., while performing a cleaning mission). The autonomous mobile robot can provide a user with information about the detected potential obstacles and/or error conditions, and receive user feedback on the detected potential obstacles and/or error conditions. The present disclosure describes various ways that user feedback on detected potential obstacles and/or error conditions can be obtained and used in operations of the robot. When the robot detects a potential obstacle and/or error condition, the robot can send information about the potential obstacle and/or error condition to a mobile computing device of the user. For example, the information can include data representing a classification of the detected potential obstacle or type of error condition, a location of the robot at the time of detecting the potential obstacle and/or error condition, and imagery captured by the robot at the time of detecting the potential obstacle and/or error condition.

(4) The mobile computing device can receive the information sent by the robot and present at least a portion of the information on a display. In some cases, the mobile computing device can solicit feedback from the user. For example, the mobile computing device can present, on the display, options that can be selected by the user to provide the feedback for controlling the autonomous mobile robot's interactions with the potential obstacle, a region where the potential obstacle is

located, or a region where the error condition was detected. For example, the user could select an option to instruct the autonomous mobile robot to clean the region (e.g., because the obstacle or the source of the error condition has been addressed by the user) or avoid the region (e.g., because the obstacle or the source of the error condition has not been or cannot be addressed by the user). The user-selectable options can also be used to provide feedback to the robot on whether or not the detected potential obstacle and/or error condition is a true obstacle and/or error condition. For example, in some cases, obstacles and/or error conditions detected by the robot are potential obstacles and/or potential error conditions that can be reviewed by the user to identify if they are true obstacles and/or true error conditions. If they are not true obstacles and/or error conditions, the user can provide feedback to denote the absence of an obstacle and/or error condition.

(5) In some examples, an autonomous mobile robot can detect one or more obstacles during a first cleaning mission, resulting in one or more areas not being cleaned by the autonomous cleaning robot during the first cleaning mission. After receiving user feedback on the one or more obstacles or areas, the robot can then perform a second cleaning mission. In some cases, the second cleaning mission can be a tidy-up mission in which the autonomous cleaning robot maneuvers relative to the one or more obstacles or areas based on the user feedback. For example, the user feedback on the one or more obstacles or areas can include a user-selected subset of the one or more areas that should be cleaned by the robot in the tidy-up mission (e.g., because an obstacle was removed or because a detected potential obstacle was not a true obstacle).

(6) Advantages of the foregoing may include, but are not limited to, those described below and herein elsewhere.

(7) Implementations described herein can improve the experience for users in interacting with autonomous mobile robots. Imagery captured by the autonomous mobile robot when an obstacle and/or error condition is detected can provide information to the user about the environment of the robot. For example, the imagery may help a user to understand why one or more areas of the environment were not cleaned by the robot during a cleaning mission. In some implementations, presenting multiple user-selectable options on a display can give the user an intuitive way to interact with and provide feedback to the robot, particularly with respect to potential obstacles and error conditions detected during a cleaning mission.

(8) Implementations described herein can also improve the coverage of the robot within the environment and the performance of the robot in future cleaning missions. For example, the plurality of user-selectable options enable the user to provide valuable user feedback to the robot. In some examples, the user feedback can provide information that is not captured by one or more sensors of the robot. The user feedback can also help a robot to process and interpret data that is captured by the one or more sensors of the robot. In some examples, the user feedback can inform the robot that an obstacle has been removed, causing the robot to clean an area during a tidy-up mission that was not cleaned during a first cleaning mission. In some examples, the user feedback can be used to create a keep-out zone, which can decrease the probability of a robot experiencing an error condition in future cleaning missions. In some examples, the user feedback can inform the robot if a detected potential obstacle is a true obstacle, which can improve the robot's detection of obstacles in future cleaning missions. In some examples, a user can opt to contribute data about potential obstacles, error conditions, and related user feedback to a database, which can be used to improve the performance of a wide range of autonomous mobile robots owned by various users.

(9) Implementations described herein can also improve the efficiency of a cleaning mission performed by the robot. For example, a tidy-up mission performed based on user feedback about potential obstacles and/or error conditions can enable the robot to efficiently clean regions that were not cleaned during a first cleaning mission while limiting re-cleaning of regions that were already cleaned during the first cleaning mission. In some cases, the regions that were not cleaned during the first cleaning mission may otherwise have gone uncleaned and may not have been cleaned in subsequent missions. The tidy-up mission can provide a mechanism to ensure that such

regions are cleaned.

(10) In one aspect, a mobile computing device includes a user input device and a controller operably connected to the user input device. The user input device includes a display, and the controller is configured to execute instructions to perform operations. The operations include presenting on the display, information about one or more areas that were not cleaned by an autonomous cleaning robot during a first mission. The operations further include transmitting data corresponding to a user-selected subset of the one or more areas to cause the autonomous cleaning robot to clean the user-selected subset during a second mission.

(11) Implementations can include one or more features below or described herein elsewhere. Implementations can include combinations of the below features.

(12) In some implementations, the information about the one or more areas can include a location of an individual one of the one or more areas. In some implementations, the information about the one or more areas can include data indicative of a potential obstacle detected in an individual one of the one or more areas. In some implementations, the data indicative of the potential obstacle detected in the individual one of the one or more areas can include one or more images of the potential obstacle, and in some implementations, the operations can further include presenting a representation of the one or more images of the potential obstacle on the display. In some implementations, the data indicative of the potential obstacle detected in the individual one of the one or more areas can include a label denoting a classification of the potential obstacle. In some implementations, the operations can further include presenting, for an individual one of the one or more areas, affordances corresponding to a plurality of user-selectable options of the display. In some implementations, the operations can further include transmitting, for the individual one of the one or more areas, data corresponding to a user-selection of one of the plurality of user-selectable options to the autonomous cleaning robot to maneuver the robot relative to the one or more areas during the second mission. In some implementations, the plurality of user-selectable options can include a first option to maneuver the autonomous cleaning robot to clean the individual one of the one or more areas and a second option to maneuver the autonomous cleaning robot to avoid the individual one of the one or more areas. In some implementations, the plurality of user-selectable options can include an option to indicate an absence of an obstacle in the individual one of the one or more areas. In some implementations, the plurality of user-selectable options can include an option to maneuver the autonomous cleaning robot to avoid the individual one of the one or more areas during the second mission, but not during one or more missions subsequent to the second mission. In some implementations, the plurality of user-selectable options can include an option to maneuver the autonomous cleaning robot to avoid the individual one of the one or more areas during the second mission and during one or more missions subsequent to the second mission. In some implementations, the affordances corresponding to the plurality of user-selectable options can be presented on the display after completion of the first mission. In some implementations, the second mission can be initiated within 12 hours of a completion of the first mission. In some implementations, the operations can further include presenting an affordance corresponding to a contribution option on the display, the contribution option enabling the mobile computing device to transmit the information about the one or more areas and the data corresponding to the user-selected subset to a database storing data from multiple users. In some implementations, the operations can further include presenting on the display, a map of an environment including the one or more areas. In some implementations, the operations can further include, presenting on the display, information about the user-selected subset after completion of the second mission. In some implementations, the first mission and the second mission can be consecutive missions. In some implementations, the second mission can include (i) the autonomous cleaning robot initiating movement from a dock to a first area of the user-selected subset, (ii) the autonomous cleaning robot initiating movement to all remaining areas of the user-selected subset, and (iii) the autonomous cleaning robot initiating movement from a last area of the user-selected subset to the dock. In some

implementations, the autonomous cleaning robot can clean the user-selected subset during the second mission without cleaning an entirety of regions cleaned by the autonomous cleaning robot during the first mission.

(13) In another aspect, an autonomous cleaning robot includes a drive system, an obstacle detection sensor, and a controller operably connected to the drive system and the obstacle detection sensor. The drive system supports the autonomous cleaning robot above a floor surface and is operable to maneuver the robot about the floor surface. The obstacle detection sensor can detect a potential obstacle as the autonomous cleaning robot is maneuvered about the floor surface, and the controller is configured to execute instructions to perform operations. The operations include performing a first mission and detecting one or more potential obstacles in one or more areas on the floor surface during the first mission. The operations further include transmitting, to a mobile computing device, data corresponding to the detected potential obstacles and the one or more areas. The operations further include receiving, from the mobile computing device, data corresponding to a user-selected subset of the one or more areas, and performing a second mission to clean the user-selected subset of the one or more areas.

(14) Implementations can include one or more features below or described herein elsewhere. Implementations can include combinations of the below features.

(15) In some implementations, the obstacle detection sensor can include an image capture device positioned on the autonomous cleaning robot to capture imagery of a portion of the floor surface forward of the autonomous cleaning robot. In some implementations, the data corresponding to the detected potential obstacles and the one or more areas can include data representing imagery of an individual one of the one or more detected potential obstacles. In some implementations, the imagery can include a single image, and in some implementations, the imagery can include a series of images. In some implementations, the data corresponding to the detected potential obstacles and the one or more areas can include a location of an individual one of the one or more areas. In some implementations, the data corresponding to the detected potential obstacles and the one or more areas can include a label denoting a classification of an individual one of the detected potential obstacles. In some implementations, the second mission can be initiated within 12 hours of a completion of the first mission. In some implementations, the operations can further include transmitting, to the mobile computing device, data corresponding to an updated status of the user-selected subset of the one or more areas during the second mission. In some implementations, the data corresponding to the updated status of the user-selected subset can include an indication of a portion of the user-selected subset that was cleaned during the second mission. In some implementations, the data corresponding to the user-selected subset of the one or more areas can include, for an individual one of the one or more areas, data corresponding to a user-selection of one of a plurality of user-selectable options. In some implementations, the plurality of user-selectable options can include a first option to cause the autonomous cleaning robot to clean the individual one of the one or more areas during the second mission and a second option to cause the autonomous cleaning robot to avoid the individual one of the one or more areas during the second mission. In some implementations, the plurality of user-selectable options can include an option to cause the autonomous cleaning robot to avoid the individual one of the one or more areas during the second mission, but not during one or more missions subsequent to the second mission. In some implementations, the plurality of user-selectable options can include an option to cause the autonomous cleaning robot to avoid the individual one of the one or more areas during the second mission and during one or more missions subsequent to the second mission. In some implementations, the plurality of user-selectable options can include an option to indicate an absence of an obstacle in the individual one of the one or more areas. In some implementations, the operations can further include updating an obstacle detection module based on the user-selection. In some implementations, the first mission and the second mission can be consecutive missions. In some implementations, performing the second mission can include (i) initiating movement from a

dock to a first area of the user-selected subset, (ii) initiating movement to all remaining areas of the user-selected subset, and (iii) initiating movement from a last area of the user-selected subset to the dock. In some implementations, the second mission can include cleaning the user-selected subset of the one or more areas without cleaning an entirety of regions cleaned by the autonomous cleaning robot during the first mission.

(16) In a further aspect, a mobile computing device includes a user input device and a controller operably connected to the user input device. The user input device includes a display, and the controller is configured to execute instructions to perform operations. The operations include receiving, from an autonomous cleaning robot, data corresponding to a detected error condition of the autonomous cleaning robot, and a portion of imagery captured by the autonomous cleaning robot. The portion of the imagery is associated with the detected error condition. The operations further include responsive to receiving the data corresponding to the detected error condition, presenting a representation of the portion of the imagery on the display and an indicator of the detected error condition.

(17) Implementations can include one or more features below or described herein elsewhere. Implementations can include combinations of the below features.

(18) In some implementations, the portion of the imagery being associated with the error condition can be captured proximate to a location of the autonomous cleaning robot when the autonomous cleaning robot detects the error condition. In some implementations, the portion of the imagery can include images captured prior to the autonomous cleaning robot detecting the error condition, and in some implementations, the portion of the imagery can include images captured subsequent to the autonomous cleaning robot detecting the error condition. In some implementations, the portion of the imagery being associated with the error condition can include a single image, and in some implementations, the portion of the imagery being associated with the error condition can include a sequence of images. In some implementations, the data corresponding to the detected error condition can include at least one of a location of the autonomous cleaning robot when the autonomous cleaning robot detects the error condition, a time when the autonomous cleaning robot detects the error condition, or a type of error condition. In some implementations, the type of error condition can be associated with a component of the autonomous cleaning robot, and the component can include at least one of a drive system, a cleaning assembly, or a brush. In some implementations, the component can be identified for replacement. In some implementations, the type of error condition can be associated with a limited mobility of the autonomous cleaning robot, and in some implementations, the limited mobility of the autonomous cleaning robot can include an inability to complete a mission or an inability to navigate to a dock. In some implementations, the portion of imagery can include imagery captured of a portion of an environment forward of the autonomous cleaning robot. In some implementations, the portion of the environment forward of the autonomous cleaning robot can include a portion of a floor surface. In some implementations, the indicator of the detected error condition can include a label of a type of error condition, and in some implementations, the indicator of the detected error condition can include a representation of a location of the autonomous cleaning robot when the autonomous cleaning robot detects the error condition.

(19) In a further aspect, an autonomous cleaning robot includes a drive system to support the autonomous cleaning robot above a floor surface, the drive system being operable to maneuver the autonomous cleaning robot about the floor surface. The autonomous cleaning robot further includes one or more sensors configured to capture sensor data corresponding to an error condition of the autonomous cleaning robot and an image capture device to capture imagery associated with the error condition. The imagery can be of a portion of an environment forward of the autonomous cleaning robot. The autonomous cleaning robot further includes one or more controllers operably connected to the drive system, the image capture device, and the one or more sensors, the one or more controllers being configured to execute instructions to perform operations. The operations

include detecting the error condition of the autonomous cleaning robot based on the sensor data as the autonomous cleaning robot is maneuvered about the floor surface. The operations further include transmitting, to a mobile computing device, (i) information about the detected error condition to cause the mobile computing device to present an indicator of the error condition and (ii) data representing a portion of the captured imagery to cause the mobile computing device to present a representation of the portion of the captured imagery.

(20) Implementations can include one or more features below or described herein elsewhere. Implementations can include combinations of the below features.

(21) In some implementations, the portion of the environment forward of the autonomous cleaning robot can include a portion of the floor surface. In some implementations, the portion of the captured imagery can be captured at a location of the autonomous cleaning robot when the autonomous cleaning robot detects the error condition. In some implementations, the portion of the captured imagery can include a single image, and in some implementations, the portion of the captured imagery can include a sequence of images. In some implementations, the information about the error condition can include at least one of a location of the autonomous cleaning robot when the autonomous cleaning robot detects the error condition, a time when the autonomous cleaning robot detects the error condition, or a type of error condition. In some implementations, the type of error condition can be associated with a component of the autonomous cleaning robot, and the component can be at least one of a drive system, a cleaning assembly, or a brush. In some implementations, the component can be identified for replacement. In some implementations, the type of error condition can be associated with a limited mobility of the autonomous cleaning robot, and in some implementations, the limited mobility of the autonomous cleaning robot can include an inability to complete a mission or an inability to navigate to a dock.

(22) The details of one or more implementations of the subject matter described in this specification are set forth in the accompanying drawings and the description below. Other potential features, aspects, and advantages will become apparent from the description, the drawings, and the claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIGS. 1A-1B are top views of an example of an environment including an autonomous mobile robot.

(2) FIG. 2 is a perspective view of an autonomous cleaning robot in an environment.

(3) FIG. 3 is a side schematic view of an autonomous cleaning robot in an environment.

(4) FIG. 4A is a bottom view of an autonomous cleaning robot.

(5) FIG. 4B is a top perspective view of the robot of FIG. 4A.

(6) FIG. 5 is a flowchart of a process of providing user feedback on obstacles detected by an autonomous mobile robot.

(7) FIG. 6 is a flowchart of a process of planning a tidy-up mission.

(8) FIG. 7 is a front view of a mobile computing device presenting a notification to plan a tidy-up mission.

(9) FIGS. 8A-8D are front views of a mobile computing device presenting a user interface for viewing obstacles detected by an autonomous cleaning robot.

(10) FIGS. 9A-9D are front views of a mobile computing device presenting a user interface for providing user feedback on potential obstacles detected by an autonomous cleaning robot.

(11) FIG. 10 is a front view of a mobile computing device presenting a user interface for initiating a tidy-up mission.

(12) FIGS. 11A-11B are front views of a mobile computing device presenting a user interface for

reviewing a completed tidy-up mission.

(13) FIG. 12 is a flowchart of a process of presenting information about an error condition detected by an autonomous cleaning robot.

(14) FIGS. 13A-13B are front views of a mobile computing device presenting a user interface for reviewing error conditions detected by an autonomous cleaning robot.

(15) FIG. 14 shows an example of a computing device and a mobile computing device.

(16) Like reference numbers and designations in the various drawings indicate like elements.

DETAILED DESCRIPTION

(17) Referring to FIG. 1A, an autonomous mobile robot **100** moves about a floor surface **10** in an environment **20**. For example, the environment **20** can be a home with a kitchen, a dining room, a bathroom, a bedroom, and a living room. The robot **100** is a cleaning robot, e.g., a robot vacuum, a robot mop, or other cleaning robot, that can clean the floor surface **10** as the robot **100** navigates about the floor surface **10**. The robot **100** can perform various missions, which can include training missions and cleaning missions. During a training mission, the robot navigates about the floor surface **10** without cleaning the floor surface **10**. During a cleaning mission, the robot **100** navigates about the floor surface **10** to clean one or more areas of the environment **20**. The areas can be determined automatically by the robot **100** or can be pre-defined by a user. At the start of the cleaning mission, the robot **100** may be docked at a docking station **50** (sometimes referred to as a dock), or may be located elsewhere in the environment **20**. A cleaning mission can be considered complete after the robot **100** has finished navigating, or attempting to navigate, to various areas in the environment **20**, when a user manually terminates the cleaning mission, or when the robot **100** experiences an error condition. In some implementations, the cleaning mission is not considered complete until the robot **100** has navigated, or attempted to navigate to the various areas a threshold number of times (e.g., twice, three times, etc.). In some implementations, the cleaning mission is not considered complete until the robot detects an amount of debris less than a threshold amount of debris in at least a portion of the various areas. The cleaning mission can also include the robot **100** returning to a docking station **50**, in the middle of the cleaning mission or at the end of the cleaning mission. In some implementations, the cleaning mission can be considered complete if the robot **100** determines that it cannot navigate to the docking station **50**. The docking station **50** can charge the robot **100**. In examples in which the robot **100** is a robot vacuum that collects debris as the robot **100** travels around the environment, in some implementations, the docking station **50** also evacuates debris from the robot **100** to allow the robot **100** to collect additional debris. Upon completion of a cleaning mission, the robot **100** stops navigating about and cleaning the floor surface **10**, until initiation of the next cleaning mission.

(18) As described in this disclosure, during a first cleaning mission, the robot **100** can detect potential obstacles in one or more areas of the environment **20**. For example, the robot **100** may detect a sock **35** as a potential obstacle in a first area **25A** of the floor surface **10**; the robot **100** may detect a cord **45** as a potential obstacle in a second area **25B** of the floor surface **10**; and the robot **100** may detect a rug **55** as a potential obstacle in a third area **25C** of the floor surface **10**. In some implementations, potential obstacles are detected by processing images captured by an image capture device of the robot **100** (e.g., a front-facing camera). For example, images captured by the image capture device of the robot **100** can be input to an object detection module **190** (shown in FIG. 4A) of the robot **100** for detecting objects within the images (e.g., using image analysis techniques, one or more convolutional neural networks, etc.). Examples of obstacle detection techniques that can be performed by, or in conjunction with, the robot **100** are described in U.S. application Ser. No. 15/863,591, filed on Jan. 5, 2018, the entire contents of which are incorporated by reference.

(19) In some implementations, to avoid getting stuck or experiencing an error condition, the robot **100** can avoid cleaning the first area **25A**, the second area **25B**, and the third area **25C** (jointly referred to as areas **25**) during the first cleaning mission. In some implementations, in response to

detecting a potential obstacle, the robot can change its behavior (e.g., shut off a vacuum, slow down, turn on a light to indicate that an obstacle has been detected, etc.). The robot **100** can send information about the areas **25**, the potential obstacles associated with the areas **25**, etc. to a mobile computing device of a user. The information can include locations of the areas **25**, an image of the associated potential obstacles, and a classification of the potential obstacles. In some cases, the robot can send information about only certain types of potential obstacles (e.g., small obstacles, obstacles that are likely to cause error conditions, obstacles easily removable from the floor surface **10** by the user, etc.). For example, the robot **100** may send information about a sock **35**, but not about a dining table **60**, even though the dining table **60** may also prevent the robot **100** from cleaning a portion of the floor surface **10**.

(20) The mobile computing device can present, on a display, affordances corresponding to user-selectable options for each of the areas **25**. The user-selectable options (further described in relation to FIGS. **9A-9D**) enable the user to provide feedback to the robot **100** about each of the areas **25** and/or the potential obstacles detected in the areas **25**. For example, if the user has picked up the sock **35**, the user can provide feedback to cause the robot to clean the area **25A** during a second cleaning mission. Alternatively, if the user is unable or unwilling to pick up the sock **35** prior to the start of a second cleaning mission, the user can provide feedback to cause the robot **100** to avoid the area **25A** during the second cleaning mission. In the case of more permanent obstacles, the user may provide feedback to the robot **100** to set up a keep-out zone. For example, if the user does not intend to move the cord **45** for a substantial period of time, the user can provide feedback to set up a keep-out zone around area **25B**, causing the robot **100** to avoid the area **25B** not only during the second cleaning mission but also during cleaning missions subsequent to the second cleaning mission. In addition, the user can provide feedback denoting the absence of an obstacle in a particular area. For example, if the robot **100** detects the rug **55** as a potential obstacle because it incorrectly identifies a pattern of the rug **55** as a cord, then the user can provide feedback to specify that there actually is no obstacle at area **25C**. This feedback can cause the robot **100** to clean the area **25C** during a second cleaning mission and can also cause an object detection module **190** (shown in FIG. **4A**) of the robot **100** to be updated (e.g. to reduce false detection of potential obstacles in future cleaning missions). In some cases, the user can choose to contribute data about each of the areas **25**, the detected potential obstacles, and/or the user feedback to a database storing data from multiple users (described in further detail herein).

(21) Referring to FIG. **1B**, after receiving feedback from the user, the robot **100** can perform a second cleaning mission. In some cases, the second cleaning mission can be a tidy-up mission. During the tidy-up mission, the robot **100** can clean a subset of the areas **25** that were not cleaned during the first cleaning mission. For example, based on the user feedback, the robot **100** can clean area **25A** and area **25C** while avoiding area **25B**. In this example, the user provides feedback to the robot **100** indicating the (i) removal of the sock **35** from area **25A**, (ii) a designation of area **25B** as a keep-out zone, and (iii) an absence of an obstacle in area **25C**. In response to receiving the user feedback, the robot **100** navigates about the floor surface **10** following a trajectory **70** during the tidy-up mission. The robot **100** starts the tidy-up mission by initiating movement from the docking station **50** to the area **25A**. The sock **35** is no longer present in the area **25A** because the user has moved it, allowing the robot **100** to clean the area **25A**. After cleaning the area **25A**, the robot **100** initiates movement to area **25C**. The user has provided feedback specifying the absence of an obstacle in the area **25C**, so the robot cleans area **25C** despite previously identifying a potential obstacle in the area **25C**. After cleaning area **25C**, the robot **100** initiates movement from the area **25C** to the docking station **50**, thereby completing the tidy-up mission. The robot **100** does not attempt to navigate to the area **25B** because the user has provided feedback that causes the area **25B** to be designated as a keep-out zone. Moreover, during the tidy-up mission, the robot cleans areas **25A** and **25C** without cleaning an entirety of regions already cleaned by the robot **100** during the first cleaning mission. Consequently, the tidy-up mission can save time and energy compared to

a second cleaning mission wherein the robot **100** attempts to clean the entire floor surface **10**.

(22) Example Autonomous Mobile Robots

(23) Examples of autonomous mobile robots are described in U.S. application Ser. No. 16/588,295, filed on Sep. 30, 2019, the entire contents of which are incorporated by reference.

(24) Referring to FIG. 2, the autonomous mobile robot **100**, e.g., an autonomous cleaning robot, on a floor surface **10** in an environment **20**, includes an image capture device **101** configured to capture imagery of the environment **20**. In particular, the image capture device **101** is positioned on a forward portion of the robot **100**. A field of view **103** of the image capture device **101** covers at least a portion of the floor surface **10** ahead of the robot **100**. The image capture device **101** can capture imagery of an object on the portion of the floor surface **10**. For example, as depicted in FIG. 2, the image capture device **101** can capture imagery representing at least a portion of a rug **30** on the floor surface **10**. The imagery can be used by the robot **100** for navigating about the environment **20** and can, in particular, be used by the robot **100** to navigate relative to the rug **30** so that the robot **100** avoids error conditions that can potentially be triggered as the robot **100** moves over the rug **30**. For example, the imagery can be used as input to an object detection module **190** (shown in FIG. 4A) that processes the imagery (e.g., executes image processing operations on data representing the imagery) to identify potential obstacles on the floor surface **10** such as clothing, electrical cords, backpacks, etc.

(25) FIGS. 3 and 4A-4B depict an example of the robot **100**. Referring to FIG. 3, the robot **100** collects debris **105** from the floor surface **10** as the robot **100** traverses the floor surface **10**. The robot **100** is usable to perform one or more cleaning missions in the environment **20** (shown in FIGS. 1A-1B) to clean the floor surface **10**. A user can provide a command to the robot **100** to initiate a cleaning mission. For example, the user can provide a start command that causes the robot **100** to initiate the cleaning mission upon receiving the start command. In another example, the user can provide a schedule that causes the robot **100** to initiate a cleaning mission at a scheduled time.

(26) Referring to FIG. 4A, the robot **100** includes a housing infrastructure **108**. The housing infrastructure **108** can define the structural periphery of the robot **100**. In some examples, the housing infrastructure **108** includes a chassis, cover, bottom plate, and bumper assembly.

(27) The robot **100** includes a drive system **110** including one or more drive wheels. The drive system **110** further includes one or more electric motors including electrically driven portions forming part of the electrical circuitry **106**. The housing infrastructure **108** supports the electrical circuitry **106**, including at least a controller **109**, within the robot **100**.

(28) The drive system **110** is operable to propel the robot **100** across the floor surface **10**. The robot **100** can be propelled in a forward drive direction F or a rearward drive direction R. The robot **100** can also be propelled such that the robot **100** turns in place or turns while moving in the forward drive direction F or the rearward drive direction R. In the example depicted in FIG. 4A, the robot **100** includes drive wheels **112** extending through a bottom portion **113** of the housing infrastructure **108**. The drive wheels **112** are rotated by motors **114** to cause movement of the robot **100** along the floor surface **10**. The robot **100** further includes a passive caster wheel **115** extending through the bottom portion **113** of the housing infrastructure **108**. The caster wheel **115** is not powered. Together, the drive wheels **112** and the caster wheel **115** cooperate to support the housing infrastructure **108** above the floor surface **10**. For example, the caster wheel **115** is disposed along a rearward portion of the housing infrastructure **108**, and the drive wheels **112** are disposed forward of the caster wheel **115**.

(29) In the example depicted in FIGS. 3, 4A, and 4B, the robot **100** is an autonomous mobile floor cleaning robot that includes a cleaning assembly **116** (shown in FIG. 4A) operable to clean the floor surface **10**. For example, the robot **100** is a vacuum cleaning robot in which the cleaning assembly **116** is operable to clean the floor surface **10** by ingesting debris **105** (shown in FIG. 3) from the floor surface **10**. The cleaning assembly **116** includes a cleaning inlet **117** through which debris is collected by the robot **100**. The cleaning assembly **116** includes one or more rotatable

members driven by a drive system (e.g., rotatable members **118** driven by a motor **120**), and the cleaning inlet **117** is positioned between the rotatable members.

(30) The rotatable members **118** are on a bottom portion of the robot **100**, and are configured to rotate to direct debris into an interior of the robot **100**, e.g., into a debris bin **124** (shown in FIG. 3). As shown in FIG. 3, the rotatable members **118** are rollers that counter-rotate relative to one another. For example, the rotatable members **118** can be rotatable about parallel horizontal axes **146**, **148** (shown in FIG. 4A) to agitate debris **105** on the floor surface **10** and direct the debris **105** toward the cleaning inlet **117**, into the cleaning inlet **117**, and into a suction pathway **145** (shown in FIG. 3) in the robot **100**.

(31) The robot **100** further includes a vacuum system **119** operable to generate an airflow through the cleaning inlet **117** between the rotatable members **118** and into the debris bin **124**. The vacuum system **119** includes an impeller and a motor to rotate the impeller to generate the airflow. The vacuum system **119** cooperates with the cleaning assembly **116** to draw debris **105** from the floor surface **10** into the debris bin **124**. In some cases, the airflow generated by the vacuum system **119** creates sufficient force to draw debris **105** on the floor surface **10** upward through the gap between the rotatable members **118** into the debris bin **124**. In some cases, the rotatable members **118** contact the floor surface **10** to agitate the debris **105** on the floor surface **10**, thereby allowing the debris **105** to be more easily ingested by the airflow generated by the vacuum system **119**.

(32) The robot **100** further includes a brush **126** that rotates about a non-horizontal axis, e.g., an axis forming an angle between 75 degrees and 90 degrees with the floor surface **10**. The robot **100** includes a motor **128** operably connected to the brush **126** to rotate the brush **126**. The brush **126** is rotatable about the non-horizontal axis in a manner that brushes debris on the floor surface **10** into a cleaning path of the cleaning assembly **116** as the robot **100** moves. The brush **126** is a side brush laterally offset from a fore-aft axis FA of the robot **100** and forwardly offset from a lateral axis LA of the robot **100** such that the brush **126** extends beyond an outer perimeter of the housing infrastructure **108** of the robot **100**. The brush **126** can thereby be capable of engaging debris on portions of the floor surface **10** that the rotatable members **118** typically cannot reach, e.g., portions of the floor surface **10** outside of a portion of the floor surface **10** directly underneath the robot **100**.

(33) The electrical circuitry **106** includes, in addition to the controller **109**, a memory storage element **144** and a sensor system with one or more electrical sensors, for example. The sensor system, as described herein, can generate a signal corresponding to a current location of the robot **100**, and can generate signals corresponding to locations of the robot **100** as the robot **100** travels along the floor surface **10**. The controller **109** is configured to execute instructions to perform one or more operations as described herein. The memory storage element **144** is accessible by the controller **109** and disposed within the housing infrastructure **108**.

(34) The one or more electrical sensors can be configured to detect features in an environment **20** of the robot **100**. For example, referring to FIG. 4A, the sensor system includes cliff sensors **134** that can detect obstacles such as drop-offs and cliffs below portions of the robot **100** where the cliff sensors **134** are disposed and redirect the robot accordingly. Referring to FIG. 4B, the sensor system further includes one or more proximity sensors (e.g., proximity sensors **136a**, **136b**) that can detect the presence or absence of objects along the floor surface **10** that are near the robot **100**. The sensor system further includes a bumper system including the bumper **138** and one or more bump sensors (e.g., bump sensors **139a**, **139b**) that detect contact between the bumper **138** and obstacles in the environment **20**. The sensor system further includes one or more obstacle following sensors (e.g., obstacle following sensor **141**) that can detect the presence or the absence of an object adjacent to a side surface of the housing infrastructure **108**. For example, the detectable objects can include obstacles such as furniture, walls, persons, and other objects in the environment **20** of the robot **100**.

(35) The sensor system can further include the image capture device **101** (shown in FIG. 4B). The image capture device **101** is positioned on a forward portion of the robot **100** and is directed to

capture imagery of at least a portion of the floor surface **10** forward of the robot **100**. In particular, the image capture device **101** can be directed in a forward direction **F** (shown in FIG. **4A**) of the robot **100**. The image capture device **101** can be, for example, a camera or an optical sensor. Referring to FIG. **2**, the field of view **103** of the image capture device **101** extends laterally and vertically. The imagery can represent portions of the floor surface **10** as well as other portions of the environment **20** above the floor surface **10**. For example, the imagery can represent portions of wall surfaces and obstacles in the environment **20** above the floor surface **10**. As described herein, the image capture device **101** can generate imagery for producing representations of maps of the environment **20** and for controlling navigation of the robot **100** about obstacles.

(36) The image capture device **101** can also be an obstacle detection sensor, wherein images captured by the image capture device **101** are used as input to an object detection module **190** (shown in FIG. **4A**) for detecting obstacles including clothing, electrical cords, backpacks, etc. (e.g., using image analysis techniques). In some implementations, the object detection module **190** can further receive, as input, data from other sensors of the robot **100** e.g., the cliff sensors **134**, the proximity sensors **136a**, **136b**, the bump sensors **139a**, **139b**, the obstacle following sensor **141**, etc., in order to detect obstacles in the environment **20**.

(37) The sensor system can further include additional obstacle detection sensors. For example, active detection technology (e.g., LIDAR, RADAR, Ultrasonic, etc.), passive detection technology, etc. can be employed instead of or in combination with the image capture device **101** to detect potential obstacles in the environment **20** and determine a distance of the potential obstacles from the robot **100**. In some implementations, the robot **100** may include a light that is used with the image capture device **101** for determining the distance of an object from the robot **100** (e.g., based on shadows, reflectivity, etc.).

(38) The sensor system can further include one or more sensors for detecting an error condition of the robot. For example, one or more electrical current sensors can provide sensor data about the electrical current provided to various components of the robot **100** including the drive wheels **112**, the cleaning assembly **116**, or the brush **126**. This sensor data can be used to detect if the robot **100** is stuck or if the cleaning assembly **116** is jammed. In some cases, the sensor data can be used to identify a component of the robot **100** (e.g., a drive wheel **112** or a brush **126**) for replacement.

(39) The sensor system can further include sensors for tracking a distance traveled by the robot **100** (e.g., motor encoders, optical sensors, etc.). The controller **109** uses data collected by the sensors of the sensor system to control navigational behaviors of the robot **100** during the mission. For example, the sensor data can be used by the controller **109** for simultaneous localization and mapping (SLAM) techniques in which the controller **109** extracts features of the environment **20** represented by the sensor data and constructs a map of the floor surface **10** of the environment **20**. The map formed from the sensor data can denote locations of traversable and non-traversable space within the environment **20**. For example, locations of obstacles are denoted on the map as non-traversable space, and locations of open floor space are denoted on the map as traversable space.

(40) The sensor data collected by any of the sensors can be stored in the memory storage element **144**. In addition, other data generated for the SLAM techniques, including mapping data, can be stored in the memory storage element **144**. These data produced during the mission can include persistent data that are produced during the mission and that are usable during a further mission. For example, the mission can be a first mission, and the further mission can be a second mission occurring after the first mission. In addition to storing the software for causing the robot **100** to perform its behaviors, the memory storage element **144** stores sensor data or data resulting from processing of the sensor data for access by the controller **109** from one mission to another mission. For example, the map is a persistent map that is usable and updateable by the controller **109** of the robot **100** from one mission to another mission to navigate the robot **100** about the floor surface **10**.

(41) The persistent data, including the persistent map, enable the robot **100** to efficiently clean the floor surface **10**. For example, the persistent map enables the controller **109** to direct the robot **100**

toward open floor space and to avoid non-traversable space. In addition, for subsequent missions, the controller **109** is able to plan navigation of the robot **100** through the environment **20** using the persistent map to optimize paths taken during the missions.

(42) The robot **100** can further include a wireless transceiver **149** (shown in FIG. **4A**). The wireless transceiver **149** allows the robot **100** to wirelessly communicate data with a communication network. The robot **100** can receive or transmit data using the wireless transceiver **149**, and can, for example, receive data representative of a map and transmit data representative of mapping data collected by the robot **100**.

(43) When the controller **109** causes the robot **100** to perform the mission, the controller **109** operates the motors **114** to drive the drive wheels **112** and propel the robot **100** along the floor surface **10**. In addition, the controller **109** operates the motor **120** to cause the rotatable members **118** to rotate, operates the motor **128** to cause the brush **126** to rotate, and operates the motor of the vacuum system **119** to generate the airflow. To cause the robot **100** to perform various navigational and cleaning behaviors, the controller **109** executes software stored on the memory storage element **144**. Execution of the software operates the various motors of the robot **100** to cause the robot **100** to perform the behaviors.

(44) Example Processes

(45) FIG. **5** illustrates a flowchart of an example process **500** of providing user feedback on obstacles detected by the autonomous mobile robot **100**. The process **500** includes operations **502**, **504**, **506**, **508**, **510**, **512**, **514**, **516**, **518**.

(46) The operations **502** and **504** involve operations of the robot **100** as the robot **100** maneuvers about the environment (e.g., environment **20** shown in FIGS. **1A-1B**). At the operation **502**, the robot **100** initiates maneuvering about the environment. For example, the robot **100** can initiate maneuvering about the environment in response to receiving a command to begin a cleaning mission. At the operation **504**, the robot **100** collects sensor data. The sensor data can be collected by a sensor system of the robot **100** and can be used for various purposes including generate mapping data of the environment and detecting potential obstacles as the robot **100** maneuvers about the environment. As the robot **100** maneuvers about the environment, the robot **100** can clean the floor surface if the operations **502** and **504** occur as part of a cleaning mission. In some cases, the robot **100** can move about the floor surface without cleaning.

(47) The operations **506**, **508**, **510** involve operations to present a user **80** with a plurality of user-selectable options associated with potential obstacles or locations of the potential obstacles in the environment. In some implementations, each user-selectable option can be presented as an affordance (e.g., a button, an interactive visual element, etc.) corresponding to the option. The operations **506** and **508** are performed by a computing system **90**, which can be a controller located on the robot **100**, a controller located on a mobile computing device **85**, a remote computing system, a distributive computing system that includes processors located on multiple devices (e.g., the robot **100**, the mobile device **85**, or a remote computing system), processors on autonomous mobile robots in addition to the robot **100**, or a combination of these computing devices. In some cases, one or more operations performed by the computing system **90** (e.g. operations **506**, **508**, **516**) can be performed by different actors. For example, detection of a potential obstacle using sensor data (operation **506**) could be performed on a remote server, while planning robot behavior relative to the potential obstacle (operation **516**) can be performed onboard the robot **100**.

(48) At the operation **506**, the sensor data is used for detecting a potential obstacle in the environment. For example, images captured by the image capture device **101** of the robot **100** can be analyzed using an object detection module **190** (e.g., that employs executable instructions of an object detection algorithm) to detect potential obstacles in the vicinity of the robot **100**. At the operation **508**, the computing system **90** generates data indicative of the detected potential obstacle in a region in the environment. For example, the data can include a portion of the imagery captured by the image capture device **101** (e.g., a single image taken at the location of the robot **100** when

the robot **100** detects the potential obstacle, a series of images, etc.), a location of the obstacle or region, and/or a label denoting a classification of the potential obstacle (e.g., a sock, a cord, a backpack, etc.). At operation **510**, the mobile computing device **85** presents affordances corresponding to a plurality of user-selectable options associated with the potential obstacle or the region to the user **80**.

(49) The user-selectable options allow the user **80** to provide user feedback to the robot **100** on the potential obstacle, and can cause the robot **100** to maneuver relative to the potential obstacle or the region in particular ways. The user-selectable options can modify various cleaning parameters of the robot **100**. For example, some user-selectable options can cause the robot **100** to perform certain actions such as cleaning a region, avoiding a region, adjusting a cleaning power of the robot in a region, etc. Some user-selectable options can modify the missions during which the robot **100** performs certain actions (e.g., an immediately subsequent mission; all subsequent missions; a specified number of subsequent missions; all subsequent missions within a day, week, month, etc.). Some user-selectable options can cause the robot **100** to include or exclude a region in a tidy-up mission, some options can modify the scheduling of a tidy-up mission (e.g., now, 30 minutes from now, 1 hour from now, 6 hours from now, 12 hours from now, etc.), and some options can modify the length of a tidy-up mission (e.g., a maximum length of 5 minutes, 15 minutes, 30 minutes, 1 hour, etc.).

(50) In some implementations, the user-selectable options can include a first option to maneuver the robot **100** to clean the region and a second option to maneuver the robot **100** to avoid the region. In some cases, multiple options can be presented to cause the robot to clean the region or avoid the region. For example, a first option can cause the robot to avoid the region only during a second cleaning mission while a second option can cause the robot to avoid the region during a second cleaning mission and in one or more cleaning missions subsequent to the second cleaning mission. The plurality of user-selectable options can also include an option to specify an absence of an obstacle in the region (e.g., if the potential obstacle detected by the robot **100** is not actually an obstacle or does not exist). In some cases, the option to specify the absence of an obstacle in the region can also cause the robot **100** to clean the region. In some cases, the option to specify the absence of an obstacle in the region can cause the robot **100** to update the object detection module **190** (e.g., a module employing a convolutional neural network) implemented by the computing system **90** in order to reduce the likelihood of false detection of potential obstacles in subsequent cleaning missions.

(51) At the operation **512**, the user **80** provides a user-selection of one of the plurality of user-selectable options. For example, based on reviewing information about the potential obstacle (e.g., one or more images of the potential obstacle, a location of the potential obstacle, a label denoting a classification of the potential obstacle, etc.), the user **80** can select one of the user-selectable options (e.g., by interacting with a corresponding affordance) to provide feedback on the potential obstacle to the robot **100**. In some cases, the information about the potential obstacle is displayed on the mobile device **85** along with the user-selectable options. At the operation **514**, the mobile device **85** can present one or more visual elements corresponding to the user-selection on a display. For example, the selected option may be shaded, outlined, changed in size, etc. to show that it has been selected.

(52) At the operation **516**, the computing system **90** plans a behavior of the robot **100** relative to the potential obstacle or the region based on the user selection. As described previously, this planned behavior can include cleaning the region or avoiding the region. Planning the behavior of the robot **100** relative to the potential obstacle can also include planning a path or trajectory of the robot to and from the region. In cases where multiple potential obstacles are detected, the planned behavior of the robot **100** can include planning a path or trajectory of the robot **100** between multiple regions associated with the multiple potential obstacles. At operation **518**, the robot **100** maneuvers relative to the obstacle or the region (e.g., in accordance with the planned robot behavior).

(53) In some implementations, multiple potential obstacles can be detected in the environment, and process **500** can be performed multiple times (e.g., once for each potential obstacle). In some cases, the user can provide feedback in real-time, providing feedback on each potential obstacle as it is detected during a mission. In other cases, the user can review multiple potential obstacles at once (e.g., after completion of a first mission), and the user-selection for each potential obstacle can be used to plan behaviors of the robot relative to the potential obstacles or regions during a second mission (e.g., a tidy-up mission).

(54) FIG. **6** illustrates a flowchart of a process **600** of planning a tidy-up mission. The operations **602** and **604** involve operations of the robot **100** as the robot **100** maneuvers about the environment (e.g., environment **20** shown in FIGS. **1A-1B**). At the operation **602**, the robot **100** performs a first mission (e.g., a cleaning mission or training mission). At the operation **604**, the robot **100** collects sensor data during the first mission. The sensor data can be collected by a sensor system of the robot **100** and can be used for various purposes including generating mapping data of the environment and detecting potential obstacles as the robot **100** maneuvers about the environment. As the robot **100** maneuvers about the environment, the robot **100** can clean the floor surface. In some cases, the robot **100** can move about the floor surface without cleaning.

(55) The operations **606**, **608**, **610** involve operations to generate information about one or more areas in the environment in which potential obstacles are detected, and to present the information to a user **80**. The operations **606** and **608** are performed by the computing system **90**, which can be a controller located on the robot **100**, a controller located on a mobile computing device **85**, a remote computing system, a distributive computing system that includes processors located on multiple devices (e.g., the robot **100**, the mobile device **85**, or a remote computing system), processors on autonomous mobile robots in addition to the robot **100**, or a combination of these computing devices. In some cases, one or more operations performed by the computing system **90** (e.g. operations **606**, **608**, **616**) can be performed by different actors. For example, detection of potential obstacles (operation **606**) could be performed on a remote server, while planning robot behavior relative to the potential obstacle (operation **616**) can be performed onboard the robot **100**.

(56) At the operation **606**, the sensor data is used for detecting one or more potential obstacles in one or more areas on the floor surface (e.g., floor surface **10** shown in FIGS. **1A-1B**) during the first mission. For example, images captured by the image capture device **101** of the robot **100** can be analyzed using the object detection module **190** (e.g., that employs executable instructions associated with an object detection algorithm) to detect potential obstacles in the vicinity of the robot **100**. At the operation **608**, the computing system **90** generates data corresponding to the detected potential obstacles and the one or more areas. For example, the data can include a portion of the imagery captured by the image capture device **101** (e.g., a single image taken at the location of the robot **100** when the robot **100** detects the potential obstacle, a series of images, a video clip, etc.), a location of the obstacle or region, and/or a label denoting a classification of the potential obstacle (e.g., a sock, a cord, a backpack, etc.). At operation **610**, the mobile computing device **85** presents information about the one or more areas to a user **80** on a display. In some implementations, the presented information can include images of the potential obstacles, locations of the potential obstacles or areas, and/or labels denoting a classification of the potential obstacles. In some implementations, affordances corresponding to a plurality of user-selectable options associated with the potential obstacles (as described in relation to FIG. **5**) can also be presented to the user **80**.

(57) At operation **612**, the user **80** provides input of a user-selected subset of the one or more areas. For example, the subset of the one or more areas can include areas that the user **80** intends for the robot **100** to clean. As similarly described in relation to FIG. **5**, in some implementations, the user selects the subset of one or more areas by interacting with an interface being presented by the mobile device **85** (e.g., options are graphically presented for selection on a display). For example, in some implementations, any of the one or more areas for which the user selects an option to cause

the robot **100** to clean the area can be included in the subset.

(58) At operation **614**, after receiving input from the user **80**, the mobile device **85** presents one or more visual elements corresponding to the user-selected subset of the one or more areas. For example, the one or more visual elements can be presented in a user interface for initiating a tidy-up mission (an example of which is shown in FIG. **10**). The one or more visual elements can include a list of the areas included in the user-selected subset, images associated with the areas included in the user-selected subset, a map representing the locations of the areas included in the user-selected subset, etc.

(59) At operation **616**, the computing system **90** plans a second mission to clean the user-selected subset of the one or more areas. Planning the second mission can include planning a path or trajectory of the robot **100** within the environment such that the robot navigates to each of the areas included in the user-selected subset. At operation **618**, the robot performs the second mission.

(60) In some implementations, the first mission and the second mission are consecutive missions. In some implementations, the second mission can be initiated within a period of time (e.g., 1 hour, 6 hours, 12 hours, 24 hours, 48 hours, etc.) after a completion of the first mission. In some implementations, during the second mission or after completion of the second mission, the robot **100** can transmit, to the mobile device **85**, data corresponding to an updated status of the user-selected subset of the one or more areas. In some implementations, the mobile computing device **85** can present at least a portion of this data on a display (e.g., described in further detail in relation to FIGS. **11A-11B**).

(61) FIGS. **7-10** are front views of the mobile computing device **85** and illustrate an example user interface (UI) for providing user feedback on obstacles detected by the autonomous mobile robot **100** (as described in relation to FIG. **5**) and planning a tidy-up mission (as described in relation to FIG. **6**). Referring to FIG. **7**, the mobile computing device **85** (sometimes referred to simply as a “mobile device”) includes a display **700**. After the robot **100** completes a cleaning mission, a push notification **710** can be sent to the mobile device **85** and presented on the display **700**. The push notification **710** can include a message stating that a cleaning mission has been completed by the robot **100** (named “Rosie” in this example), but that some areas were not cleaned during the cleaning mission. The push notification **710** can encourage the user to review these areas for future cleanings. In some implementations, reviewing the areas includes providing user feedback on the areas and potential obstacles detected by the robot **100** in those areas. In some implementations, reviewing the areas can include planning a tidy-up mission.

(62) Referring to FIG. **8A**, after receiving the push notification **710**, the user **80** can navigate to a screen presented on the display **700** (e.g., by interacting with the push notification, by opening a mobile application, etc.) for viewing obstacles detected by the robot **100**. The display **700** can include a time stamp **822** that includes a day and time when the cleaning mission was started or when the cleaning mission was completed. The display **700** can also include a map **802** of an environment in which the robot **100** performed the cleaning mission. In this example, the environment is a home including a den, a dining room, a kitchen, a living room, and a bedroom (not shown). The display **700** also includes control options to interact with the map **802** including a zoom-in option **806A**, a zoom-out option **806B**, and a rotate map option **808** that rotates the map (e.g., by 90 degrees in a clockwise direction). Indicators **804A**, **804B**, **804C**, **804D** (jointly referred to as indicators **804**) are presented on the map **802**, showing the location of potential obstacles detected by the robot **100** during the cleaning mission. In FIG. **8A**, the indicator **804A** is selected, as depicted by a change in color and enlarged size. While the indicators **804** are shown as exclamation points, this is not intended to be limiting, and other symbols can be used. For example, in some implementations, the indicators **804** may include symbols corresponding to a type of the potential obstacle (e.g., a sock, a shoe, a cord, etc.).

(63) The display **700** also includes a clean report **810**, which presents additional information about the one or more potential obstacles detected by the robot **100**. The clean report **810** can include a

summary statement **812** which provides information about a total number of potential obstacles detected in the environment during the cleaning mission (e.g., **5** obstacles in this example). For each potential obstacle, the clean report **810** can include an image of the potential obstacle (e.g., images **814A**, **814B**), a label denoting a classification of the potential obstacle (e.g., labels **816A**, **816B**), and an area where the potential obstacle was detected (e.g., areas **818A**, **818B**). Referring now to FIG. **8D**, in some implementations, the clean report **810** can be a textual report. The clean report **810** can include a title **830** for the mission, which can specify an environment **20** the robot **100** is in (e.g., “Main Floor”) or a type of mission performed by the robot **100** (e.g., “Cleaning Mission”, “Training Mission”, etc.). In some cases, the title **830** can be edited and customized by the user **80**. The clean report **810** can also include various entries (e.g., entry **832**, entry **840**) that are listed in chronological order to create a timeline of the mission. Entry **832** is an example of an entry for a particular area in the environment (e.g., environment **20**), and includes a status indicator **834**, a name of the area **836**, and a time spent **838** in the area. In this example, entry **832** shows that a kitchen was successfully cleaned for 24 minutes. Entry **840** is an example of an entry for a potential obstacle detected in the environment, and includes a status indicator **842** (which may be different from the status indicator **834**), a message **844**, and a time that the potential obstacle was encountered **846**. In some implementations, the message **844** can state that a potential obstacle was detected, and in some implementations, the message **844** can include information about the obstacle such as a classification and a location of the potential obstacle (e.g., “Socks in Dining Room”). In this example, entry **840** shows that socks were detected in the dining room at 9:42 AM. In some implementations, the entry **840** can include a user-selectable affordance **890** that, when selected, can cause the robot **100** to clean an area associated with the location of the potential obstacle during a subsequent mission (e.g., a tidy-up mission). In some implementations imagery associated with the potential obstacle can be included in the clean report **810** (e.g., as part of the entry **840**, next to the entry **840**, at the bottom of the clean report **810**, etc.). In some implementations, the entry **840** can be selected by the user **80** (e.g., by touching the entry **840**, touching and holding the entry **840**, hovering a cursor over the entry **840**, etc.) in order to view further information about the potential obstacle including imagery associated with the potential obstacle and/or a map representing the location of the potential obstacle within the environment. In some implementations, the clean report **810** can further include a summary entry **895** that can include, for example, a date of the mission, a time of the mission, and/or a time spent to complete the mission. In some implementations, only a portion of the information described may be presented within the clean report **810**, and in some implementations additional information can be presented.

(64) In some implementations, the clean report **810** can be displayed prior to the user **80** reviewing further information about each potential obstacle (described in further detail with respect to FIGS. **9A-9D**). In an example of such implementations, if the user selects affordance **890** from the clean report **810**, no further review of the potential obstacle may be required in order to initiate a subsequent mission (e.g., a tidy-up mission). In another example, selection of affordance **890** can cause the robot to immediately return to the area associated with a potential obstacle without requiring feedback about other potential obstacles. In some implementations, the clean report **810** can be displayed in real-time as a mission is being performed, and selection of affordance **890** can cause the robot **100** to return to the area associated with a potential obstacle during the same mission.

(65) In some implementations, the clean report **810** can be displayed after the user **80** has already reviewed further information about each potential obstacle and provided feedback about the potential obstacles. In such cases, a user-selection of affordance **890** from the clean report **810** can adjust or override previously provided feedback about a potential obstacle. For example, subsequent to user review of the potential obstacles, the clean report **810** can serve as a user interface for planning a tidy-up mission. Examples of user interfaces for planning tidy-up missions

are described in further detail herein with respect to FIG. 10.

(66) Referring again to FIG. 8A, because indicator **804A** is selected, information about a potential obstacle associated with indicator **804A** is displayed in a primary position within the clean report **810**. In this example, the potential obstacle associated with the indicator **804A** is a cord located in the den. In some implementations, multiple potential obstacles are presented simultaneously in the clean report **810**. For example, the potential obstacles can be presented in an order corresponding to the chronological order during which the robot **100** detected the potential obstacles. In this example, the next potential obstacle detected by the robot **100** was a group of socks in the dining room, as shown by image **814B**, label **816B**, and area **818B**.

(67) In order to view information about the remainder of the detected potential obstacles, the user **80** can select any of the indicators **804** to view the potential obstacles corresponding to those indicators. Alternatively, the user **80** can scroll (e.g., using a left-to-right or right-to-left gesture on the display **700**) through each of the potential obstacles detected by the robot **100**. For example, referring to FIG. 8B, in response to the user selecting indicator **804B**, information about the potential obstacle associated with indicator **804B** (e.g., socks in the dining room) is displayed in the primary position within the clean report **810**. In this example, the next potential obstacle detected after the socks in the dining room was clothing in the bedroom, as shown by image **814C**, label **816C**, and area **818C**. In this example, the map **802** is partially covered by the clean report **810**, so that the bedroom is not visible to the user **80**. In some implementations, a minimization option **824** can be selected by the user **80** to minimize the clean report **810** in order to view a larger portion of the map **802**.

(68) Referring now to FIG. 8C, the user can scroll through the clean report **810** to view a remainder of the potential obstacles detected by the robot **100** during the cleaning mission. In this example, the user has scrolled through the potential obstacles such that an image **814D**, label **816D**, and area **818D** (corresponding to a backpack in the living room) are positioned in the primary position of the clean report **810**. In response to the user's scrolling, the indicator **804C**, corresponding to the backpack in the living room, is depicted by a change in color and enlarged size as though it had been directly selected by the user **80**. In this example, the next potential obstacle detected after the backpack in the living room was a cord in the living room, as shown by image **814E**, label **816E**, and area **818E**.

(69) At any point while viewing the potential obstacles detected by the robot **100**, the user **80** can select an option **820** in order to begin providing user feedback on the potential obstacles. After selecting the option **820**, the user **80** is presented with a series of screens on the display **700** to provide feedback on the potential obstacles. FIGS. 9A-9D illustrate examples of such screens. Referring to FIG. 9A, the user interface for providing user feedback on potential obstacles includes information about the potential obstacle including an image **902A** of the potential obstacle, a label **908A** denoting a classification of the potential obstacle (e.g., a cord **904**), and an area **910A** of the potential obstacle (e.g., a den). In some implementations, the image **902A** may be a single image captured at a location of the robot **100** when the robot detects the potential obstacles (e.g., using image capture device **101** shown in FIG. 2). In some implementations, a series of images can be presented, wherein the series of images are captured in a vicinity of a location of the potential obstacle. In some implementations, a bounding box **906A** may be presented with the image **902A** in order to highlight a specific portion of the image **902A** where a potential obstacle has been detected (e.g. via an object detection module **190** employing a convolutional neural network). In some implementations, an area of the image **902A** that is not included inside the bounding box **906A** can be darkened to further highlight the specific portion of the image **902A** where the potential obstacle has been detected. In some implementations, the user interface for providing user feedback on potential obstacles can include additional information such as a confidence score or accuracy metric associated with the label **908A**.

(70) In addition to information about the potential obstacle (e.g., the cord **904**), the display **700** can

include a plurality of user-selectable affordances **912A**, **912B**, **912C**, **912D** (collectively referred to as user-selectable affordances **912**) associated with the potential obstacle or an area in the environment where the potential obstacle was detected. In some implementations, each of the affordances **912** can correspond to a user-selectable option. For example, an affordance **912A**, if selected by the user **80**, can cause the robot **100** to clean the area (e.g., because the obstacle has been removed by the user **80**) during an immediately subsequent mission (e.g., a tidy-up mission) and/or during another future mission (e.g., an upcoming scheduled mission). The affordance **912B** can correspond to the absence of an obstacle in the area (e.g., because the user **80** has determined that the potential obstacle detected by the robot **100** is not an obstacle and/or that the area was clear). Selection of affordance **912B** can cause the robot **100** to clean the area during an immediately subsequent mission and/or during another future mission, and can further cause the robot **100** to update the object detection module **190** to reduce the likelihood of false detection of potential obstacles in future cleaning missions. Affordance **912C** can cause the robot **100** to avoid the area during an immediately subsequent cleaning mission (e.g., during a tidy-up mission), but not during additional cleaning missions. Affordance **912D** can cause the robot **100** to avoid the area during an immediately subsequent cleaning mission (e.g., during a tidy-up mission) and during additional cleaning missions. For example, selection of affordance **912D** can cause the establishment of a keep-out zone around the area, preventing the robot **100** from cleaning the area until the keep-out zone is removed or edited by the user **80**. When the user selects an affordance from the plurality of user-selectable affordances **912**, visual element corresponding to the user-selection can be presented on the display **700**. For example, the selected affordance may appear with a shaded background, a colored background, bold text, a bold outline (as shown with affordance **912B** in FIG. **9D**), a colored outline, etc. While the affordances **912** are shown with particular icons, this is not intended to be limiting, and other icons and/or symbols can be used.

(71) Referring again to FIG. **9A**, the display **700** can further include an affordance **930** corresponding to a user-selectable contribution option. If the user selects the affordance **930** (denoted by the display of a checkmark icon), the mobile device **85** can transmit information about the potential obstacle (e.g., data representing image **902A**, label **908A**, and/or area **910A**) and the user-selection associated with the potential obstacle (e.g., one of the plurality of user-selectable affordances **912**) to a database storing data from multiple users. In some implementations, this data may be used for improving the object detection module **190** of various autonomous mobile robots owned by different users. In some implementations, this data may be used for the development of new autonomous mobile robots. In some cases, the default setting for the affordance **930** is to not transmit information to the database. In some implementations, a user can control what information is transmitted to the database (e.g., transmitting information corresponding to the label **908A** and the user-selection, but not transmitting information corresponding to the image **902A**). In some implementations, images that contain people or human faces can be automatically filtered and prevented from being transmitted to the database. In some implementations, a user can crop the image **902A** (e.g., by reshaping the bounding box **906A**) in order to crop out sensitive data prior to transmitting information to the database. In some implementations, the display **700** can include a user-selectable option **914** that, when selected, can cause the user **80** to be presented with further information about what data is transmitted to the database and what the data is used for.

(72) The display **700** can further include one or more visual elements that represent the number of potential obstacles detected during the cleaning mission as well as the potential obstacle currently being reviewed. For example, in FIG. **9A**, the visual elements **916**, **990** show that five potential obstacles are available to be reviewed by the user **80**, and the user **80** is currently viewing a first of the five potential obstacles. In visual element **916**, this is shown by the first of five circles being filled. In visual element **990**, this is shown by displaying a number corresponding to the potential obstacle currently being reviewed (“1” in this example) as well as a number corresponding to the total number of potential obstacles available to be reviewed (“5” in this example). In some

implementations, the user **80** can scroll through the potential obstacles for review by using a left-to-right or right-to-left swiping motion on the display **700**.

(73) In FIG. **9A**, the cord **904** has been detected as a potential obstacle by the robot **100**. If the cord is unlikely to be removed prior to the start of the next cleaning mission, the user **80** can select affordance **912C** or affordance **912D** to cause the robot to avoid the cord **904**. If affordance **912C** is selected, the robot **100** will avoid the cord **904** during an immediately subsequent cleaning mission (e.g., a tidy-up mission), but not during additional cleaning missions after that. However, if affordance **912D** is selected, the robot **100** will avoid the cord **904** in both an immediately subsequent cleaning mission (e.g., a tidy-up mission) and additional cleaning missions after that. Thus, in scenarios where the user **80** does not expect the cord **904** to be removed for a substantial period of time, the user may choose to select affordance **912D** instead of affordance **912C**. After selecting an affordance from the plurality of user-selectable affordances **912**, the user **80** can then provide feedback on another potential obstacle detected by the robot **100**.

(74) Referring now to FIG. **9B**, a group of socks **918** has been detected as a potential obstacle by the robot **100**. The display **700** of FIG. **9B** is similar to the display **700** of FIG. **9A**, but presents a different image **902B**, a different bounding box **906B**, a different label **908B**, and a different area **910B** associated with the potential obstacle. In this example, the bounding box **906B** is presented with the image **902B** to highlight a portion of the image **902B** in which a group of socks **918** was detected. If the user is able to remove the group of socks **918** prior to the start of the next cleaning mission, the user can select affordance **912A** to cause the robot **100** to clean the area associated with the socks **918** during a subsequent cleaning mission (e.g., during a tidy-up mission). Alternatively, if the user is unable or unwilling to remove the group of socks **918** prior to the start of the next cleaning mission, the user can select affordance **912C** to cause the robot to avoid the area associated with the socks **918** in an immediately subsequent cleaning mission (e.g., a tidy-up mission). In this example, if the user **80** intends to eventually pick up the socks **918**, the user **80** may decide not to select affordance **912D**, since the socks **918** are only a temporary obstacle in the environment. After selecting an affordance from the user-selectable affordances **912**, the user **80** can then provide feedback on another potential obstacle detected by the robot **100**.

(75) Referring now to FIG. **9C**, a shadow **920** on the floor surface **950** has been detected as a potential obstacle by the robot **100**. The display **700** of FIG. **9C** is similar to the display **700** of FIGS. **9A** and **9B**, but presents a different image **902C**, a different bounding box **906C**, a different label **908C**, and a different area **910C** associated with the potential obstacle. In this example, the bounding box **906C** is presented with the image **902C** to highlight a portion of the image **902C** in which a potential obstacle has been detected. In this example, the robot **100** has mistakenly identified a shadow **920** on the floor surface **950** as a cord (as shown by label **908C**). After reviewing the information presented on the display **700**, the user **80** can select affordance **912B** to specify the absence of an obstacle in the area. Selection of affordance **912B** can cause the robot **100** to update the object detection module **190** to reduce false detection of potential obstacles in the future and cause the robot **100** to clean the area during a subsequent cleaning mission (e.g., a tidy-up mission). After selecting an affordance from the user-selectable affordances **912**, the user **80** can then provide feedback on another potential obstacle detected by the robot **100**. Referring to FIG. **9D**, if the shadow **920** is the last potential obstacle requiring feedback from the user, an affordance **924** can be presented on the display to allow the user **80** to finish reviewing the potential obstacles detected by the robot **100**. If the user **80** is not finished reviewing the potential obstacles detected by the robot **100**, the user **80** can scroll through the potential obstacles and edit previously-made user-selections.

(76) Upon selection of the affordance **924**, a screen can be presented on the display **700**, providing a user interface for planning a tidy-up mission. Referring to FIG. **10**, the user interface for planning a tidy-up mission can include a “Re-Clean” portion **1000A** including visual elements corresponding to a user-selected subset of areas to be cleaned during the tidy-up mission. For example, the visual

elements can include images of the areas such as image **1002A**, image **1002B**, and image **1002C** (jointly referred to as images **1002**). The user-selected subset of areas can correspond to areas where the user **80** had previously selected an affordance to cause the robot **100** to clean the area while reviewing information about a corresponding potential obstacle. The images **1002** can each include a selection indicator such as a checkmark to allow the user **80** to confirm whether or not an area associated with each image should be cleaned during the tidy-up mission. For example, image **1002A** can include selection indicator **1004A** to confirm that the user **80** wants the area associated with the socks in the dining room to be cleaned during the tidy-up mission. The image **1002B** can include selection indicator **1004B** to confirm that the user **80** wants the area associated with the clothes in the bedroom to be cleaned during the tidy-up mission. The image **1002C** can include selection indicator **1004C** to confirm that the user **80** wants the area associated with the misidentified cord in the living room to be cleaned during the tidy-up mission. If the user **80** does not wish to include an area for re-cleaning during the tidy-up mission, the user **80** can touch the corresponding image in the “Re-Clean” section **1000A** to toggle the selection indicator and deselect the image.

(77) The user interface for planning a tidy-up mission can further include, on the display **700**, an “Avoid This Time” portion **1000B**, and/or an “Avoid Always” portion (not shown). Each portion can include visual elements corresponding to a subset of areas based on a previous user-selection for each area from the plurality of user-selectable affordances **912** (shown in FIGS. **9A-9D**). The user **80** can scroll through the “Re-Clean” portion **1000A**, the “Avoid This Time” portion **1000B**, and the “Avoid Always” portion (not shown) by performing a top-to-bottom or bottom-to-top swiping motion on the display **700** in order to view the contents of each portion. In some implementations, the user **80** can drag-and-drop a visual element (e.g., image **1002A**) from one portion of the display **700** to another portion of the display **700** (e.g., from a “Re-Clean” portion **1000A** to an “Avoid This Time” portion **1000B**). In some implementations, the user interface for planning a tidy-up mission can further allow for the user **80** to provide a cleaning order for the areas included for re-cleaning in the tidy-up mission.

(78) The user interface for planning a tidy-up mission can further include a message **1010** and corresponding affordances **1006**, **1008** to prompt the user **80** to set a start time for the tidy-up mission. If the user selects affordance **1008**, the robot **100** will initiate the tidy-up mission right away. Alternatively, if the user **80** selects affordance **1006**, the user **80** will be presented with a user-interface for scheduling the tidy-up mission at a later time.

(79) Based on the user input, the robot **100** plans and performs a tidy-up mission. Upon completion of the tidy-up mission, the mobile computing device **85** can present on the display **700**, a user interface for reviewing the completed tidy-up mission. Referring to FIG. **11A**, the user interface for reviewing the completed tidy-up mission can include, on the display **700**, a time stamp **1122** that includes a day and time when the tidy-up mission was started or when the tidy-up mission was completed. The display **700** can also include a map **1102** of an environment in which the robot **100** performed the tidy-up mission. In this example, the environment is a home including a den, a dining room, a kitchen, a living room, and a bedroom (not shown). The display **700** also includes control options to interact with the map **1102** including a zoom-in option **1106A**, a zoom-out option **1106B**, and a rotate map option **1108** that rotates the map (e.g., by 90 degrees in a clockwise direction). Indicators **1104A**, **1104B**, **1104C**, **1104D**, **1104E** (jointly referred to as indicators **1104**) are presented on the map **1102**. Indicator **1104A** is a shaded area that indicates that the user **80** has instructed the robot **100** to avoid the corresponding area of the environment. For example, the user may have designated a keep-out zone around a previously detected cord in the den. Indicators **1104B**, **1104C**, **1104D** are checkmarks that indicate areas that were previously avoided by the robot **100** during a first cleaning mission, but successfully cleaned by the robot **100** during the tidy-up mission. For example, these areas may include an area where the user **80** removed a potential obstacle (e.g., socks in the dining room) or an area where the user **80** provided feedback specifying

the absence of an obstacle (e.g., the absence of a falsely detected cord in the living room). During the tidy-up mission, the robot **100** may encounter one or more new potential obstacles that were not detected during the first cleaning mission. For example, indicator **1104E** is an exclamation point indicating a newly detected potential obstacle (e.g., a cord in the kitchen). In FIG. **11A**, the indicator **1104E** is selected, as depicted by a change in color and an enlarged size of the indicator **1104E**. The display **700** can further include, on the map **1102**, a path **1128** traversed by the robot **100** during the tidy-up mission. In some implementations, the path **1128** traversed by the robot **100** during the tidy-up mission can be planned to avoid covering an entirety of regions navigated to by the robot **100** during the first cleaning mission. In some implementations, the path **1128** can be planned by the robot **100** to cover all the areas included in the tidy-up mission in a time and energy efficient manner. In some implementations, the user **80** can provide the robot **100** with a cleaning order for the areas included in the tidy-up mission, which may assist in planning of the path **1128**.

(80) The user interface for reviewing the completed tidy-up mission further includes, on the display **700**, a clean report **1110**, which presents additional information about any new potential obstacles detected by the robot **100**. Similar to the clean report **810** (shown in FIG. **8A**), the clean report **1110** can include a summary statement **1112** which provides information about a total number of new potential obstacles detected in the environment during the tidy-up mission (e.g., **1** new obstacle in this example). For each potential obstacle, the clean report **1110** can include an image of the potential obstacle (e.g., image **1114**), a label denoting a classification of the potential obstacle (e.g., label **1116**), and an area where the potential obstacle was detected (e.g., area **1118**). In some implementations, a minimization option **1124** can be selected by the user **80** to minimize the clean report **1110** in order to view a larger portion of the map **1102**.

(81) Referring now to FIG. **11B**, in some implementations, the clean report **1110** can be a textual report. The clean report **1110** can include a portion **1130** corresponding to a first mission performed by the robot **100**. For example, the portion **1130** can be substantially similar to the textual clean report **810** described in relation to FIG. **8D**. The clean report **1110** can further include a portion **1142** corresponding to the completed tidy-up mission. In this example, the portion **1142** includes entries **1132**, **1134** listed in chronological order to create a timeline of the mission. Using entry **1134** as an example, each entry in the portion **1142** can include a status indicator **1136**, an area **1138**, and a time spent **1140** by the robot **100** in the area. In this example, the robot **100** has completed a tidy-up mission by re-cleaning the living room for 5 minutes (i.e., where a cord was detected during the first mission) and the dining room for 7 minutes (i.e., where socks were detected during the first mission). In some implementations, the time spent by the robot **100** in an area during the tidy-up mission can be less than a time spent in the area during the first mission because only a portion of the area is cleaned during the tidy-up mission (e.g., only a portion of the living room where a cord was previously detected, only a portion of the dining room where socks were previously detected, etc.). In some implementations, each entry of the clean report **1110** (e.g., entry **1134**) can be selected by the user **80** in order to view further information about the entry (e.g., a user-selected affordance that caused the robot **100** to perform a cleaning task, a map displaying a location of the area, etc.). In some implementations, only a portion of the information described herein may be presented within the clean report **1110**, and in some implementations, additional information can be presented. For example, in addition to information about areas corresponding to previously detected potential obstacles, the clean report **1110** may also include information about keep-out zones.

(82) The user interface for reviewing the completed tidy-up mission further includes, on the display **700**, an affordance **1120** to finish the tidy-up mission (shown in both FIG. **11A** and FIG. **11B**). In some implementations, after the completion of a first tidy-up mission, the user **80** may not be able to provide feedback on newly detected potential obstacles and/or may not be able to schedule a second consecutive tidy-up mission. Upon finishing review of the tidy-up mission, the user **80** can

select the affordance **1120** to exit from the user interface for reviewing the completed tidy-up mission.

(83) While examples have been provided for the detection of potential obstacles by the robot **100**, in some implementations, the robot **100** may detect error conditions during a cleaning mission. Various types of error conditions can be detected. For example, an error condition can be associated with a component of the autonomous cleaning robot **100** including the drive system **110** of the robot **100**, a cleaning assembly **116** of the robot **100**, or a brush **126** of the robot **100**. For example, one or more of the components may be broken or jammed. In some implementations, detecting the error condition can include identifying a component of the robot **100** for replacement. In some implementations, the type of error condition can be associated with a limited mobility of the robot **100**. For example, the robot **100** can be stuck underneath a piece of furniture such as a sofa. In another example, the robot **100** may be able to drive about the floor surface, but cannot find a path to escape from a certain portion of the environment. In another example, the robot **100** may be unable to navigate to a docking station **50** or may be unable to complete a cleaning mission.

(84) FIG. **12** illustrates a process **1200** of presenting information about an error condition detected by the robot **100**. The process **1200** includes operations **1202**, **1204**, **1206**, **1208**, **1210**, **1212**, **1214**.

(85) The operations **1202** and **1204** involve operations of the robot **100** as the robot **100** maneuvers about the environment (e.g., environment **20** shown in FIGS. **1A-1B**). At the operation **1202**, the robot **100** initiates maneuvering about the environment. For example, the robot **100** can initiate maneuvering about the environment in response to receiving a command to begin a cleaning mission. At the operation **1204**, the robot **100** collects sensor data. The sensor data can be collected by a sensor system of the robot **100** and can be used for various purposes including generating mapping data of the environment and detecting error conditions as the robot **100** maneuvers about the environment. As the robot **100** maneuvers about the environment, the robot **100** can clean the floor surface if the operations **1202** and **1204** occur as part of a cleaning mission. In some cases, the robot **100** can move about the floor surface without cleaning.

(86) The operations **1206**, **1208**, **1210** involve operations to detect an error condition, generate data corresponding to the detected error condition, and obtain a portion of imagery associated with the error condition. The operations **1206**, **1208**, **1210** are performed by a computing system **90**, which can be a controller located on the robot **100**, a controller located on a mobile computing device **85**, a remote computing system, a distributive computing system that includes processors located on multiple devices (e.g., the robot **100**, the mobile device **85**, or a remote computing system), processors on autonomous mobile robots in addition to the robot **100**, or a combination of these computing devices. In some cases, one or more operations performed by the computing system **90** (e.g. operations **1206**, **1208**, **1210**) can be performed by different actors. For example, detection of error conditions (operation **1206**) could be performed onboard the robot **100**, while obtaining a portion of imagery associated with the error condition (operation **1210**) can be performed on a remote server. At the operation **1206**, the sensor data is used for detecting an error condition of the robot **100**. For example, data from an electrical current sensor can be used to detect if a drive system **110**, a cleaning assembly **116**, or a brush **126** of the robot **100** is stuck. In some examples, sensor data used for SLAM techniques can be used to detect that the robot **100** is unable to escape a certain portion of the environment or navigate to a docking station **50**. At operation **1208**, the computing system **90** generates data corresponding to the detected error condition. For example, the data can include a location of the robot **100** when the error condition is detected and/or a label denoting a type of the error condition. At operation **1210**, the computing system **90** obtains a portion of imagery associated with the error condition. For example, the computing system **90** can obtain the portion of imagery from images captured by the image capture device **101**. The portion of the imagery associated with the error condition can be a single image, a sequence of images, or a video clip. In some implementations, the portion of the imagery can be captured proximate to a location of the robot **100** when the robot **100** detects the error condition. In some implementations,

the portion of the imagery can be captured prior to the robot **100** detecting the error condition. In some implementations, the portion of the imagery can be captured subsequent to the robot **100** detecting the error condition.

(87) At operation **1212**, the mobile device **85** presents a representation of the portion of the imagery and an indicator of the detected error condition. For example, as shown in FIGS. **13A-13B** (described in further detail herein), the mobile device **85** can present an image associated with the error condition (e.g., images **1305**, **1315**), a label denoting a type of the error condition (e.g., labels **1312**, **1320**), and/or a location of the error condition (e.g., indicators **1304**, **1318**).

(88) At operation **1214**, the user **80** reviews the representation of the portion of the imagery and the indicator of the detected error condition. In some implementations, the user may provide feedback about the detected error condition. For example, the user **80** can provide a user-selection from the mobile computing device **85** that causes the robot **100** to shut off and/or terminate a cleaning mission. In some implementations, the user **80** is not given an option to provide feedback about the detected error condition. In such cases, the user **80** is still able to review the representation of the portion of the imagery and the indicator of the detected error condition in order to gain context for the error condition detected by the robot **100**. In some cases, the user **80** can opt to contribute data about the error condition to a database that stores data from multiple users. This data can be used to reduce the number of error conditions experienced by existing autonomous mobile robots and to develop new autonomous mobile robots.

(89) Referring to FIG. **13A**, an example of a user interface for reviewing an error condition detected by the robot **100** is presented on the display **700** of the mobile computing device **85**. The display **700** can include a time stamp **1322** that includes a day and time when the error condition was detected. The display **700** can also include a map **1302** of an environment in which the robot **100** is located. In this example, the environment is a home including a den, a dining room, a kitchen, a living room, and a bedroom (not shown). The display **700** also includes control options to interact with the map **1302** including a zoom-in option **1306A**, a zoom-out option **1306B**, and a rotate map option **1308** that rotates the map (e.g., by 90 degrees in a clockwise direction). An indicator **1304** is presented on the map **1302**, indicating the location of the robot **100** when the error condition is detected during the cleaning mission. The display **700** further includes an error report **1310**. The error report **1310** can include a label **1312** denoting a type of error condition. In the example of FIG. **13A**, the label **1312** indicates that the robot **100** (named “Rosie”) is stuck. The error report can also include an image **1305**, which is a portion of the imagery captured by the robot **100**. For example, the image **1305** could have been captured at, or proximal to, a location of the robot **100** when the robot **100** detected the error condition. In this example, the image **1305** depicts a cord **1314** on the floor surface. The user **80** can review the items presented on the display **700** and gain context regarding the error condition detected by the robot **100**. For example, the user **80** can infer that the robot **100** is stuck in the living room of the home because it has become tangled up by electrical cord **1314**.

(90) In some implementations, the image **1305** could have been captured by the robot **100** within a period of time (e.g., 1 s, 5 s, 10 s, etc.) prior to detecting the error condition or within a period of time (e.g., 1 s, 5 s, 10 s, etc.) subsequent to detecting the error condition. In some implementations, the user **80** can view multiple images taken at various times prior to and/or subsequent to detection of the error condition. This can be advantageous when an initial image (e.g., image **1305**) is not sufficient for the user **80** to gain context regarding the error condition. For example, if the robot **100** becomes stuck underneath a couch and detects an error condition, images taken thereafter may be too dark or may lack features that would allow the user **80** to contextualize the error condition. However, an image taken prior to detection of the error condition could reveal to the user **80** that the robot **100** was heading toward a space underneath the couch before it became stuck. In another example, if the robot **100** detects that it cannot navigate to the docking station **50**, an initial image **1305** may not include enough features for the user **80** to determine the current location of the robot

100. However, an image taken subsequent to detection of the error condition (e.g., 5 s subsequent to detection of the error condition) may include features (e.g., a television, rug, potted plant, etc.) that reveal to the user **80** that the robot **100** is currently in the living room.

(91) In some implementations, the user **80** can select an affordance **1316** corresponding to a contribution option to contribute information about the error condition to a database that stores data from multiple users. For example, if the affordance **1316** is selected, the mobile computing device **85** can transmit information about the error condition (e.g., a location of the error condition, sensor data corresponding to the error condition, a portion of the imagery captured by the robot **100**, a label denoting a type of the error condition, etc.) to the database. This data can be used to reduce the number of error conditions experienced by the robot **100** in future cleaning mission or to develop new autonomous mobile robots.

(92) Referring to FIG. **13B**, another example of a user interface for reviewing an error condition detected by the robot **100** is presented on the display **700** of the mobile computing device **85**. This example is similar to the example described in relation to FIG. **13A**, with similar elements labeled with the same reference numerals. An indicator **1318** is presented on the map **1302**, indicating the location of the robot **100** when the error condition is detected during the cleaning mission. In this case, the error report **1310** includes a label **1320** denoting a type of the error condition. In the example of FIG. **13B**, the label **1320** indicates that a cleaning assembly **116** of the robot **100** is jammed. The error report can also include an image **1315**, which is a portion of the imagery captured by the robot **100**. In this example, the image **1315** depicts a pile of clothes **1324** on the floor surface. The user **80** can review the items presented on the display **700** and gain context regarding the error condition detected by the robot **100**. For example, the user **80** can infer that a cleaning assembly **116** of the robot **100** became jammed because it attempted to vacuum a piece of clothing (e.g., a sock) from the pile of clothes **1324** located in the den. Once again, the user can select an affordance **1316** corresponding to a contribution option to contribute information about the error condition to a database that stores data from multiple users. For example, if the affordance **1316** is selected, the mobile computing device **85** can transmit information about the error condition (e.g., a location of the error condition, sensor data corresponding to the error condition, a portion of the imagery captured by the robot **100**, a label denoting a type of the error condition, etc.) to the database. This data can be used to reduce the number of error conditions experienced by the robot **100** in future cleaning mission or to develop new autonomous mobile robots.

(93) Further Alternative Implementations

(94) Although a few implementations have been described in detail above, other implementations are possible. For example, in some implementations, there may be a time limit (e.g., 12 hours from completion of a first mission, 18 hours from completion of a first mission, 24 hours from completion of a first mission, etc.) for providing user feedback on detected obstacles and/or error conditions. In some implementations, if the user **80** does not provide feedback on the detected obstacles and/or error conditions with the time limit, the information about the detected obstacles and/or error conditions can be deleted. In some implementations, once the time limit expires, the user **80** may no longer be able to plan a tidy-up mission or cause the robot **100** to initiate a tidy-up mission. In some implementations, prior to expiration of the time limit, a notification (e.g., a push notification) can be presented on the display **700** of the mobile computing device **85** to remind the user to provide feedback on potential obstacles and/or error conditions detected by the robot **100**.

(95) In some implementations, a first mission performed by the robot **100** may be a training mission, in which the robot navigates about the environment **20** without cleaning the floor surface **10**. During the training mission, the robot **100** can still detect potential obstacles and/or error conditions (e.g., in a manner substantially similar to the examples described above). Based on user feedback on the potential obstacles and/or error conditions, the robot **100** can then perform a second mission. In some cases, the second mission may be a cleaning mission such as a tidy-up mission.

(96) In some implementations, information about one or more potential obstacles and/or error conditions detected by the robot **100** can be shared between multiple autonomous mobile robots (e.g., a fleet of autonomous mobile robots) owned by the user **80**. In some implementations, if one robot detects an error condition and is unable to complete a cleaning mission, a second robot can complete the cleaning mission. In some implementations, a first robot can perform a first cleaning mission, and a second robot can perform a tidy-up mission after receiving feedback on one or more potential obstacles from the user **80**.

(97) In some implementations, prior to the user **80** providing feedback on potential obstacles and/or error conditions detected by the robot **100**, the mobile computing device **85** can present on the display **700**, a user interface for a tutorial. The tutorial can explain to the user **80** how to provide feedback on potential obstacles and/or error conditions detected by the robot **100**. The tutorial can also explain how to plan a tidy-up mission. The tutorial can provide information about each of the plurality of user-selectable affordances **912** (shown in FIGS. **9A-9D**) as well as detail about the affordances **930**, **1316** corresponding to a contribution option (shown in FIGS. **9A-9D**, **13A-13B**). In some implementations, the tutorial is only shown prior to a first attempt by the user **80** to provide feedback on potential obstacles and/or error conditions detected by the robot **100**. However, in some implementations, the tutorial can be accessed at any point in time so that the user **80** can review the tutorial as needed.

(98) FIG. **14** shows an example of a computing device **1400** and a mobile computing device **1450** that can be used to implement the techniques described here. For example, the computing device **1400** and the mobile computing device **1450** can represent an example of the mobile device **85** and elements of the computing system **90**. The computing device **1400** is intended to represent various forms of digital computers, such as laptops, desktops, workstations, personal digital assistants, servers, blade servers, mainframes, and other appropriate computers. The mobile computing device **1450** is intended to represent various forms of mobile devices, such as personal digital assistants, cellular telephones, smart-phones, and other similar computing devices. Additionally, computing device **1400** or **1450** can include Universal Serial Bus (USB) flash drives. The USB flash drives may store operating systems and other applications. The USB flash drives can include input/output components, such as a wireless transmitter or USB connector that may be inserted into a USB port of another computing device. The components shown here, their connections and relationships, and their functions, are meant to be examples only, and are not meant to be limiting.

(99) The computing device **1400** includes a processor **1402**, a memory **1404**, a storage device **1406**, a high-speed interface **1408** connecting to the memory **1404** and multiple high-speed expansion ports **1410**, and a low-speed interface **1412** connecting to a low-speed expansion port **1414** and the storage device **1406**. Each of the processor **1402**, the memory **1404**, the storage device **1406**, the high-speed interface **1408**, the high-speed expansion ports **1410**, and the low-speed interface **1412**, are interconnected using various busses, and may be mounted on a common motherboard or in other manners as appropriate. The processor **1402** can process instructions for execution within the computing device **1400**, including instructions stored in the memory **1404** or on the storage device **1406** to display graphical information for a GUI on an external input/output device, such as a display **1416** coupled to the high-speed interface **1408**. In other implementations, multiple processors and/or multiple buses may be used, as appropriate, along with multiple memories and types of memory. Also, multiple computing devices may be connected, with each device providing portions of the necessary operations (e.g., as a server bank, a group of blade servers, or a multi-processor system).

(100) The memory **1404** stores information within the computing device **1400**. In some implementations, the memory **1404** is a volatile memory unit or units. In some implementations, the memory **1404** is a non-volatile memory unit or units. The memory **1404** may also be another form of computer-readable medium, such as a magnetic or optical disk.

(101) The storage device **1406** is capable of providing mass storage for the computing device **1400**.

In some implementations, the storage device **1406** may be or contain a computer-readable medium, such as a floppy disk device, a hard disk device, an optical disk device, or a tape device, a flash memory or other similar solid state memory device, or an array of devices, including devices in a storage area network or other configurations. Instructions can be stored in an information carrier. The instructions, when executed by one or more processing devices (for example, processor **1402**), perform one or more methods, such as those described above. The instructions can also be stored by one or more storage devices such as computer- or machine-readable mediums (for example, the memory **1404**, the storage device **1406**, or memory on the processor **1402**).

(102) The high-speed interface **1408** manages bandwidth-intensive operations for the computing device **1400**, while the low-speed interface **1412** manages lower bandwidth-intensive operations. Such allocation of functions is an example only. In some implementations, the high-speed interface **1408** is coupled to the memory **1404**, the display **1416** (e.g., through a graphics processor or accelerator), and to the high-speed expansion ports **1410**, which may accept various expansion cards. In the implementation, the low-speed interface **1412** is coupled to the storage device **1406** and the low-speed expansion port **1414**. The low-speed expansion port **1414**, which may include various communication ports (e.g., USB, Bluetooth, Ethernet, wireless Ethernet) may be coupled to one or more input/output devices. Such input/output devices may include a scanner **1430**, a printing device **1434**, or a keyboard or mouse **1436**. The input/output devices may also be coupled to the low-speed expansion port **1414** through a network adapter. Such network input/output devices may include, for example, a switch or router **1432**.

(103) The computing device **1400** may be implemented in a number of different forms, as shown in FIG. **14**. For example, it may be implemented as a standard server **1420**, or multiple times in a group of such servers. In addition, it may be implemented in a personal computer such as a laptop computer **1422**. It may also be implemented as part of a rack server system **1424**. Alternatively, components from the computing device **1400** may be combined with other components in a mobile device, such as a mobile computing device **1450**. Each of such devices may contain one or more of the computing device **1400** and the mobile computing device **1450**, and an entire system may be made up of multiple computing devices communicating with each other.

(104) The mobile computing device **1450** includes a processor **1452**, a memory **1464**, an input/output device such as a display **1454**, a communication interface **1466**, and a transceiver **1468**, among other components. The mobile computing device **1450** may also be provided with a storage device, such as a micro-drive or other device, to provide additional storage. Each of the processor **1452**, the memory **1464**, the display **1454**, the communication interface **1466**, and the transceiver **1468**, are interconnected using various buses, and several of the components may be mounted on a common motherboard or in other manners as appropriate.

(105) The processor **1452** can execute instructions within the mobile computing device **1450**, including instructions stored in the memory **1464**. The processor **1452** may be implemented as a chipset of chips that include separate and multiple analog and digital processors. For example, the processor **1452** may be a Complex Instruction Set Computers (CISC) processor, a Reduced Instruction Set Computer (RISC) processor, or a Minimal Instruction Set Computer (MISC) processor. The processor **1452** may provide, for example, for coordination of the other components of the mobile computing device **1450**, such as control of user interfaces, applications run by the mobile computing device **1450**, and wireless communication by the mobile computing device **1450**.

(106) The processor **1452** may communicate with a user through a control interface **1458** and a display interface **1456** coupled to the display **1454**. The display **1454** may be, for example, a Thin-Film-Transistor Liquid Crystal Display (TFT) display or an Organic Light Emitting Diode (OLED) display, or other appropriate display technology. The display interface **1456** may comprise appropriate circuitry for driving the display **1454** to present graphical and other information to a user. The control interface **1458** may receive commands from a user and convert them for

submission to the processor **1452**. In addition, an external interface **1462** may provide communication with the processor **1452**, so as to enable near area communication of the mobile computing device **1450** with other devices. The external interface **1462** may provide, for example, for wired communication in some implementations, or for wireless communication in other implementations, and multiple interfaces may also be used.

(107) The memory **1464** stores information within the mobile computing device **1450**. The memory **1464** can be implemented as one or more of a computer-readable medium or media, a volatile memory unit or units, or a non-volatile memory unit or units. An expansion memory **1474** may also be provided and connected to the mobile computing device **1450** through an expansion interface **1472**, which may include, for example, a Single In-Line Memory Module (SIMM) card interface. The expansion memory **1474** may provide extra storage space for the mobile computing device **1450**, or may also store applications or other information for the mobile computing device **1450**. Specifically, the expansion memory **1474** may include instructions to carry out or supplement the processes described above, and may include secure information also. Thus, for example, the expansion memory **1474** may be provided as a security module for the mobile computing device **1450**, and may be programmed with instructions that permit secure use of the mobile computing device **1450**. In addition, secure applications may be provided via the SIMM cards, along with additional information, such as placing identifying information on the SIMM card in a non-hackable manner.

(108) The memory may include, for example, flash memory and/or non-volatile random access memory (NVRAM), as discussed below. In some implementations, instructions are stored in an information carrier. The instructions, when executed by one or more processing devices (for example, processor **1452**), perform one or more methods, such as those described above. The instructions can also be stored by one or more storage devices, such as one or more computer- or machine-readable mediums (for example, the memory **1464**, the expansion memory **1474**, or memory on the processor **1452**). In some implementations, the instructions can be received in a propagated signal, for example, over the transceiver **1468** or the external interface **1462**.

(109) The mobile computing device **1450** may communicate wirelessly through the communication interface **1466**, which may include digital signal processing circuitry where necessary. The communication interface **1466** may provide for communications under various modes or protocols, such as Global System for Mobile communications (GSM) voice calls, Short Message Service (SMS), Enhanced Messaging Service (EMS), or Multimedia Messaging Service (MMS) messaging, code division multiple access (CDMA), time division multiple access (TDMA), Personal Digital Cellular (PDC), Wideband Code Division Multiple Access (WCDMA), CDMA2000, or General Packet Radio Service (GPRS), among others. Such communication may occur, for example, through the transceiver **1468** using a radio-frequency. In addition, short-range communication may occur, such as using a Bluetooth, Wi-Fi, or other such transceiver. In addition, a Global Positioning System (GPS) receiver module **1470** may provide additional navigation- and location-related wireless data to the mobile computing device **1450**, which may be used as appropriate by applications running on the mobile computing device **1450**. In some implementations, the wireless transceiver **149** of the robot **100** can employ any of the wireless transmission techniques provided for by the communication interface **1466** (e.g., to communicate with the mobile device **85**).

(110) The mobile computing device **1450** may also communicate audibly using an audio codec **1460**, which may receive spoken information from a user and convert it to usable digital information. The audio codec **1460** may likewise generate audible sound for a user, such as through a speaker, e.g., in a handset of the mobile computing device **1450**. Such sound may include sound from voice telephone calls, may include recorded sound (e.g., voice messages, music files, etc.) and may also include sound generated by applications operating on the mobile computing device **1450**.

(111) The mobile computing device **1450** may be implemented in a number of different forms, as

shown in the figure. For example, it may be implemented as a cellular telephone **1480**. It may also be implemented as part of a smart-phone, personal digital assistant **1482**, or other similar mobile device.

(112) Various implementations of the systems and techniques described here can be realized in digital electronic circuitry, integrated circuitry, specially designed application specific integrated circuits (ASICs), computer hardware, firmware, software, and/or combinations thereof. These various implementations can include implementation in one or more computer programs that are executable and/or interpretable on a programmable system including at least one programmable processor, which may be special or general purpose, coupled to receive data and instructions from, and to transmit data and instructions to, a storage system, at least one input device, and at least one output device.

(113) These computer programs (also known as programs, software, software applications or code) include machine instructions for a programmable processor, and can be implemented in a high-level procedural and/or object-oriented programming language, and/or in assembly/machine language. As used herein, the terms machine-readable medium and computer-readable medium refer to any computer program product, apparatus and/or device (e.g., magnetic discs, optical disks, memory, Programmable Logic Devices (PLDs)) used to provide machine instructions and/or data to a programmable processor, including a machine-readable medium that receives machine instructions as a machine-readable signal. The term machine-readable signal refers to any signal used to provide machine instructions and/or data to a programmable processor. In some implementations, modules (e.g., an object detection module), functions (e.g., presenting information on a display), and processes executed by the robot **100**, the computing system **90**, and the mobile device **85** (described in relation to FIG. 5) can execute instructions associated with the computer programs described above.

(114) To provide for interaction with a user, the systems and techniques described here can be implemented on a computer having a display device (e.g., a cathode ray tube (CRT) or liquid crystal display (LCD) monitor) for displaying information to the user and a keyboard and a pointing device (e.g., a mouse or a trackball) by which the user can provide input to the computer. Other kinds of devices can be used to provide for interaction with a user as well; for example, feedback provided to the user can be any form of sensory feedback (e.g., visual feedback, auditory feedback, or tactile feedback); and input from the user can be received in any form, including acoustic, speech, or tactile input.

(115) The systems and techniques described here can be implemented in a computing system that includes a back end component (e.g., as a data server), or that includes a middleware component (e.g., an application server), or that includes a front end component (e.g., a client computer having a graphical user interface or a Web browser through which a user can interact with an implementation of the systems and techniques described here), or any combination of such back end, middleware, or front end components. The components of the system can be interconnected by any form or medium of digital data communication (e.g., a communication network). Examples of communication networks include a local area network (LAN), a wide area network (WAN), and the Internet.

(116) The computing system can include clients and servers. A client and server are generally remote from each other and typically interact through a communication network. The relationship of client and server arises by virtue of computer programs running on the respective computers and having a client-server relationship to each other.

Claims

1. A mobile computing device comprising: a user input device comprising a display; and a controller operably connected to the user input device, the controller configured to execute

instructions to perform operations comprising: presenting on the display, (i) information about one or more areas that were not cleaned by an autonomous cleaning robot during a first mission, wherein the information comprises an image of a first potential obstacle of one or more potential obstacles, the image captured by the autonomous cleaning robot, (ii) an indication that the one or more areas were not cleaned by the autonomous cleaning robot in response to a determination of a presence of the one or more potential obstacles in the one or more areas, the determination being made by an analysis of imagery of the one or more areas captured by the autonomous cleaning robot, wherein the first potential obstacle is in a first area of the one or more areas, and (iii) at least one affordance selectable by a user to control whether the autonomous cleaning robot will clean the first area in a second mission, wherein the affordance is presented specifically in response to identification of the first potential obstacle as a potential obstacle, and based on user interaction with the at least one affordance, transmitting data to control whether the autonomous cleaning robot will clean the first area in the second mission.

2. The mobile computing device of claim 1, wherein the information presented on the display comprises an identifier of a room in which the first potential obstacle is located.

3. The mobile computing device of claim 1, wherein the information presented on the display comprises a boundary overlaid on the image, the boundary indicating the first potential obstacle in the image.

4. The mobile computing device of claim 1, wherein a portion of the image that does not include the first potential obstacle is darkened compared to a portion of the image that includes the first potential obstacle, to highlight the first potential obstacle in the image.

5. The mobile computing device of claim 1, wherein the information presented on the display further comprises, alongside the image, a second image of a second potential obstacle of the one or more potential obstacles, and wherein the operations further comprise presenting an affordance corresponding to a user-selectable option to review the first potential obstacle and the second potential obstacle one-by-one in sequence.

6. The mobile computing device of claim 1, wherein the information presented on the display comprises a label denoting a classification of the first potential obstacle.

7. The mobile computing device of claim 1, wherein the at least one affordance comprises a first affordance that, when selected, causes data to be sent to the autonomous cleaning robot indicating that the first potential obstacle is not an obstacle.

8. The mobile computing device of claim 7, wherein the first affordance, when selected, causes data to be transmitted to control the autonomous cleaning robot to update an object detection module of the autonomous cleaning robot based on the first potential obstacle not being an obstacle.

9. The mobile computing device of claim 1, wherein the at least one affordance comprises: a first affordance that, when selected, causes data to be transmitted to control the autonomous cleaning robot to avoid the first area during an immediately subsequent mission and not avoid the first area during future missions after the immediately subsequent mission, and a second affordance that, when selected, causes data to be transmitted to control the autonomous cleaning robot to avoid the first area during the immediately subsequent mission and during the future missions.

10. The mobile computing device of claim 1, wherein the operations further comprise presenting an affordance corresponding to a contribution option on the display, the contribution option enabling the mobile computing device to transmit the information about the one or more areas and a result of the user interaction with the at least one affordance to a database storing data from multiple users.

11. The mobile computing device of claim 1, wherein the operations further comprise, presenting on the display, information about whether the one or more areas were cleaned by the autonomous cleaning robot during the second mission.

12. The mobile computing device of claim 1, wherein the data causes the autonomous cleaning

robot to clean a user-selected subset of the one or more areas during the second mission without cleaning an entirety of regions cleaned by the autonomous cleaning robot during the first mission.
